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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 1st February Evening Shift

JEE Main Question Paper with Answer Key

1 Feb 2023	90	17	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	23 questions
Physics	28 questions
Out of syllabus	17 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

Open solution

$O-O$ bond length in H_2O_2 is $\underline{X}$ than the $O-O$ bond length in F_2O_2. The $O-H$ bond length in H_2O_2 is $\underline{Y}$ than that of the $O-F$ bond in F_2O_2.

Choose the correct option for $\underline{X}$ and $\underline{Y}$ from those given below :

(A) X - shorter, Y - longer

(B) X - longer, Y - shorter

(C) X - shorter, Y - shorter

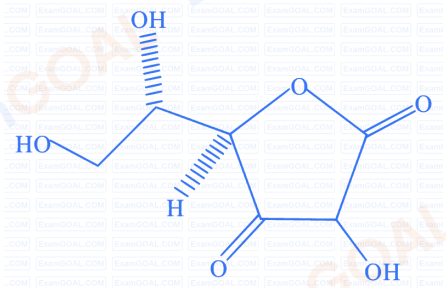
(D) X - longer, Y - longer

Q2. MCQ (Single Correct Answer)

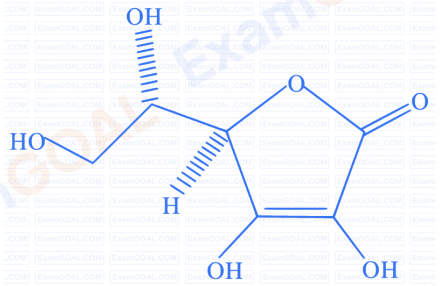
Open solution

All structures given below are of vitamin C. Most stable of them is :

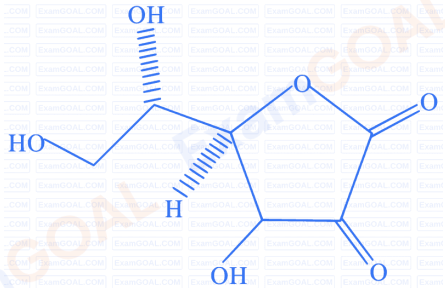
(A)



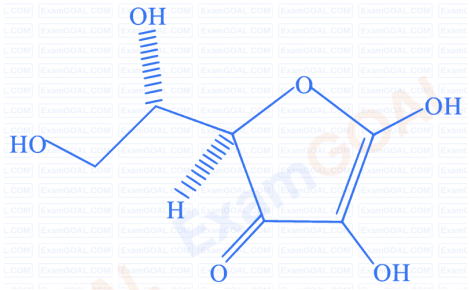
(B)



(C)



(D)



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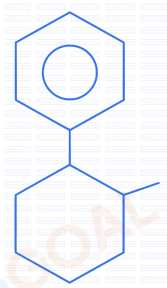
Q3. MCQ (Single Correct Answer)

'X' is :

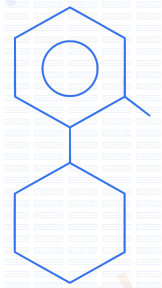


Open solution

(A)



(B)



(C)



(D)

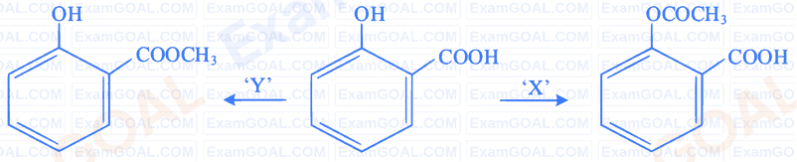


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Q4. MCQ (Single Correct Answer)

Open solution

In a reaction,



reagents 'X' and 'Y' respectively are :

(A) $CH_3OH/H^+, \Delta$ and $(CH_3CO)_2O/H^+$

(B) $(CH_3CO)_2O/H^+$ and $CH_3OH/H^+, \Delta$

(C) $CH_3OH/H^+, \Delta$ and $CH_3OH/H^+, \Delta$

(D) $(CH_3CO)_2O/H^+$ and $(CH_3CO)_2O/H^+$

Q5. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Sulphanilic acid gives esterification test for carboxyl group.

Statement II : Sulphanilic acid gives red colour in Lassaigne's test for extra element detection.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Statement I is correct but Statement II is incorrect

(C) Both Statement I and Statement II are incorrect

(D) Statement I is incorrect but Statement II is correct

Q6. MCQ (Single Correct Answer)

Open solution

Which element is not present in Nessler's reagent?

(A) Mercury (B) Iodine (C) Potassium (D) Oxygen

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Q7. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : α-halocarboxylic acid on reaction with dil. NH_3 gives good yield of α-amino carboxylic acid whereas the yield of amines is very low when prepared from alkyl halides.

Reason (R) : Amino acids exist in zwitter ion form in aqueous medium.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is not correct but (R) is correct

(B) (A) is correct but (R) is not correct

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Q8. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Cu^{2+} in water is more stable than Cu^{+}.

Reason (R) : Enthalpy of hydration for Cu^{2+} is much less than that of Cu^{+}.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is correct but (R) is not correct

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) (A) is not correct but (R) is correct

(D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Q9. MCQ (More than One Correct Answer)

Open solution

The effect of addition of helium gas to the following reaction in equilibrium state, is :

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(A) the equilibrium will shift in the forward direction and more of Cl_2 and PCl_3 gases will be produced.

(B) addition of helium will not affect the equilibrium.

(C) helium will deactivate PCl_5 and reaction will stop.

(D) the equilibrium will go backward due to suppression of dissociation of PCl_5.

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Q10. MCQ (Single Correct Answer)

Open solution

Which one of the following sets of ions represents a collection of isoelectronic species?

(Given : Atomic Number : $\mathrm{F}: 9, \mathrm{Cl}: 17, \mathrm{Na}=11, \mathrm{Mg}=12, \mathrm{Al}=13, \mathrm{K}=19, \mathrm{Ca}=20, \mathrm{Sc}=21$)

(A) $\mathrm{N}^{3-}, \mathrm{O}^{2-}, \mathrm{F}^{-}, \mathrm{S}^{2-}$

(B) $\mathrm{Ba}^{2+}, \mathrm{Sr}^{2+}, \mathrm{K}^{+}, \mathrm{Ca}^{2+}$

(C) $\mathrm{Li}^{+}, \mathrm{Na}^{+}, \mathrm{Mg}^{2+}, \mathrm{Ca}^{2+}$

(D) $\mathrm{K}^{+}, \mathrm{Cl}^{-}, \mathrm{Ca}^{2+}, \mathrm{Sc}^{3+}$

Q11. MCQ (Single Correct Answer)

Open solution

For electron gain enthalpies of the elements denoted as $\Delta_{\mathrm{eg}}\mathrm{H}$, the incorrect option is :

(A) $\Delta_{\mathrm{eg}}\mathrm{H}(\mathrm{I}) < \Delta_{\mathrm{eg}}\mathrm{H}(\mathrm{At})$

(B) $\Delta_{\mathrm{eg}}\mathrm{H}(\mathrm{Te}) < \Delta_{\mathrm{eg}}\mathrm{H}(\mathrm{Po})$

(C) $\Delta_{\mathrm{eg}}\mathrm{H}(\mathrm{Cl}) < \Delta_{\mathrm{eg}}\mathrm{H}(\mathrm{F})$

(D) $\Delta_{\mathrm{eg}}\mathrm{H}(\mathrm{Se}) < \Delta_{\mathrm{eg}}\mathrm{H}(\mathrm{S})$

Q12. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : An aqueous solution of KOH when used for volumetric analysis, its concentration should be checked before the use.

Reason (R) : On aging, KOH solution absorbs atmospheric CO_2.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is correct but (R) is not correct

(B) (A) is not correct but (R) is correct

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Q13. MCQ (Single Correct Answer)

Open solution

The complex cation which has two isomers is :

(A) $[\mathrm{Co}(\mathrm{NH}_3)_5\mathrm{Cl}]^{+}$

(B) $[\mathrm{Co}(\mathrm{H}_2\mathrm{O})_6]^{3+}$

(C) $[\mathrm{Co}(\mathrm{NH}_3)_5\mathrm{NO}_2]^{2+}$

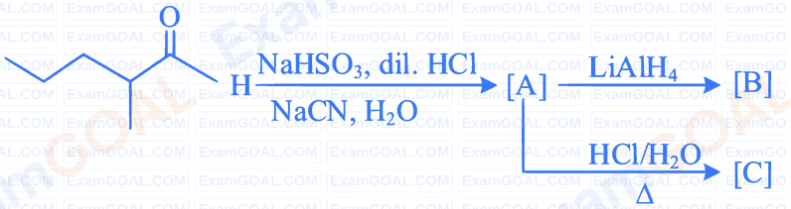
(D) $[\mathrm{Co}(\mathrm{NH}_3)_5\mathrm{Cl}]^{2+}$

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Q14. MCQ (Single Correct Answer)

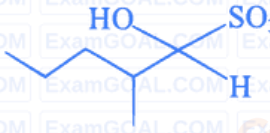
Open solution

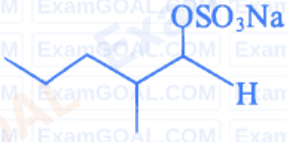
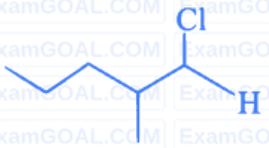
The structures of major products A, B and C in the following reaction sequence are :



(A) A =  , B =  , C = 

(B) A =  , B =  , C = 

(C) A =  , B =  , C = 

(D) A =  , B =  , C = 

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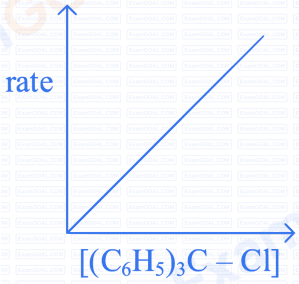
Q15. MCQ (Single Correct Answer)

Open solution

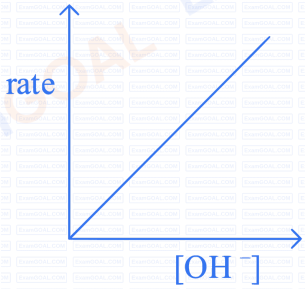
The graph which represents the following reaction is :

$$(C_6H_5)_3C-Cl \xrightarrow[\text{Pyridine}]{OH^-} (C_6H_5)_3C-OH$$

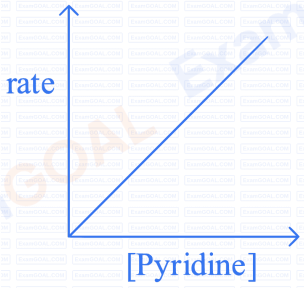
(A)



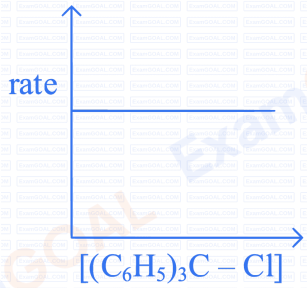
(B)



(C)



(D)



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Q16. Numerical

1×10^{-5} M $AgNO_3$ is added to 1 L of saturated solution of AgBr. The conductivity of this solution at 298 K is _____ $\times 10^{-8}$ S m^{-1}. (Nearest integer)

[Given : $K_{SP}(AgBr) = 4.9 \times 10^{-13}$ at 298 K

$\lambda^0_{Ag^+} = 6 \times 10^{-3}$ S m^2 mol^{-1}

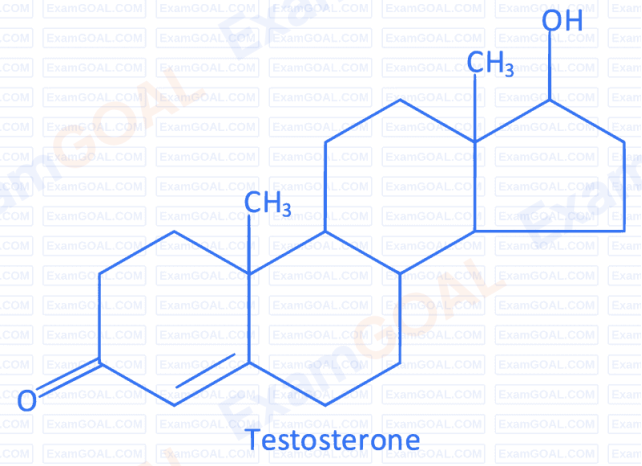
$\lambda^0_{Br^-} = 8 \times 10^{-3}$ S m^2 mol^{-1}

$\lambda^0_{NO_3^-} = 7 \times 10^{-3}$ S m^2 mol^{-1}]

Open solution

Q17. Numerical

Testosterone, which is a steroidal hormone, has the following structure.



Testosterone

The total number of asymmetric carbon atom/s in testosterone is _____.

Open solution

Q18. Numerical

20% of acetic acid is dissociated when its 5 g is added to 500 mL of water. The depression in freezing point of such water is _____ $\times 10^{-3}$°C.

Atomic mass of C, H and O are 12,1 and 16 a.m.u. respectively.

[Given : Molal depression constant and density of water are 1.86 K kg mol^{-1} and 1 g cm^{-3} respectively.]

Open solution

Q19. Numerical

0.3 g of ethane undergoes combustion at 27°C in a bomb calorimeter. The temperature of calorimeter system (including the water) is found to rise by 0.5°C. The heat evolved during combustion of ethane at constant pressure is _____ kJ mol^{-1}. (Nearest integer)

[Given : The heat capacity of the calorimeter system is 20 kJ K^{-1}, R = 8.3 JK^{-1} mol^{-1}.

Assume ideal gas behaviour.

Atomic mass of C and H are 12 and 1 g mol^{-1} respectively]

Open solution

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Q20. Numerical

The molality of a $10\%(\mathrm{v/v})$ solution of di-bromine solution in $\mathrm{CCl_4}$ (carbon tetrachloride) is 'x'. $x =$ _____ $\times 10^{-2}$ m. (Nearest integer)

[Given : molar mass of $\mathrm{Br_2} = 160\ \mathrm{g\ mol^{-1}}$

atomic mass of $\mathrm{C} = 12\ \mathrm{g\ mol^{-1}}$

atomic mass of $\mathrm{Cl} = 35.5\ \mathrm{g\ mol^{-1}}$

density of dibromine $= 3.2\ \mathrm{g\ cm^{-3}}$

density of $\mathrm{CCl_4} = 1.6\ \mathrm{g\ cm^{-3}}$]

Open solution

Q21. Numerical

The spin only magnetic moment of $[\mathrm{Mn(H_2O)_6}]^{2+}$ complexes is _____ B.M. (Nearest integer)

(Given : Atomic no. of Mn is 25)

Open solution

Q22. Numerical

$\mathrm{A} \rightarrow \mathrm{B}$

The above reaction is of zero order. Half life of this reaction is $50\ \mathrm{min}$. The time taken for the concentration of A to reduce to one-fourth of its initial value is _____ min. (Nearest integer)

Open solution

Mathematics 23 questions

Q23. MCQ (Single Correct Answer)

Let $9 = x_1 < x_2 < \ldots < x_7$ be in an A.P. with common difference d. If the standard deviation of $x_1, x_2 \ldots, x_7$ is 4 and the mean is $\bar{x}$, then $\bar{x} + x_6$ is equal to :

(A) $2\left(9 + \frac{8}{\sqrt{7}}\right)$

(B) 25

(C) $18\left(1 + \frac{1}{\sqrt{3}}\right)$

(D) 34

Open solution

Q24. MCQ (Single Correct Answer)

Let $P(S)$ denote the power set of $S = \{1, 2, 3, \ldots, 10\}$. Define the relations R_1 and R_2 on $P(S)$ as $\mathrm{AR_1\ B}$ if $(\mathrm{A} \cap \mathrm{B^c}) \cup (\mathrm{B} \cap \mathrm{A^c}) = \emptyset$ and $\mathrm{AR_2\ B}$ if $\mathrm{A} \cup \mathrm{B^c} = \mathrm{B} \cup \mathrm{A^c}, \forall \mathrm{A}, \mathrm{B} \in \mathrm{P(S)}$. Then :

(A) only R_2 is an equivalence relation

(B) both R_1 and R_2 are not equivalence relations

(C) both R_1 and R_2 are equivalence relations

(D) only R_1 is an equivalence relation

Open solution

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Q25. MCQ (Single Correct Answer)

The sum of the absolute maximum and minimum values of the function $f(x) = |x^2 - 5x + 6| - 3x + 2$ in the interval $[-1, 3]$ is equal to :

(A) 13 (B) 24 (C) 10 (D) 12

Open solution

Q26. MCQ (Single Correct Answer)

Let $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + 5\hat{k}$ be two vectors. Then which one of the following statements is TRUE ?

(A) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-13}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(B) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{13}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(C) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{13}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

(D) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-13}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

Open solution

Q27. MCQ (Single Correct Answer)

For the system of linear equations $\alpha x + y + z = 1, x + \alpha y + z = 1, x + y + \alpha z = \beta$, which one of the following statements is **NOT** correct?

(A) It has infinitely many solutions if $\alpha = 1$ and $\beta = 1$

(B) It has infinitely many solutions if $\alpha = 2$ and $\beta = -1$

(C) $x + y + z = \frac{3}{4}$ if $\alpha = 2$ and $\beta = 1$

(D) It has no solution if $\alpha = -2$ and $\beta = 1$

Open solution

Q28. MCQ (Single Correct Answer)

If $y(x) = x^x, x > 0$, then $y''(2) - 2y'(2)$ is equal to

(A) $4(\log_e 2)^2 + 2$ (B) $8\log_e 2 - 2$ (C) $4\log_e 2 + 2$ (D) $4(\log_e 2)^2 - 2$

Open solution

Q29. MCQ (Single Correct Answer)

The value of the integral

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x+\frac{\pi}{4}}{2-\cos 2x} dx$ is :

(A) $\frac{\pi^2}{6\sqrt{3}}$ (B) $\frac{\pi^2}{6}$ (C) $\frac{\pi^2}{3\sqrt{3}}$ (D) $\frac{\pi^2}{12\sqrt{3}}$

Open solution

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Q30. MCQ (Single Correct Answer)

The area of the region given by $\{(x, y) : xy \leq 8, 1 \leq y \leq x^2\}$ is :

(A) $16\log_e 2 - \frac{14}{3}$ (B) $8\log_e 2 - \frac{13}{3}$ (C) $16\log_e 2 + \frac{7}{3}$ (D) $8\log_e 2 + \frac{7}{6}$

Open solution

Q31. MCQ (Single Correct Answer)

The number of integral values of k, for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval (1, 2) and its other root lies in the interval (2, 3), is :

(A) 2 (B) 0 (C) 1 (D) 3

Open solution

Q32. MCQ (Single Correct Answer)

Let a, b be two real numbers such that $ab < 0$. If the complex number $\frac{1+ai}{b+i}$ is of unit modulus and $a + ib$ lies on the circle $|z - 1| = |2z|$, then a possible value of $\frac{1+[a]}{4b}$, where $[t]$ is greatest integer function, is :

(A) $\left(\frac{1+\sqrt{7}}{4}\right)$ (B) $\frac{1}{2}$

(C) 0 (D) -1

Open solution

Q33. MCQ (Single Correct Answer)

If $A = \frac{1}{2}\begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$, then :

(A) $A^{30} - A^{25} = 2I$ (B) $A^{30} + A^{25} - A = I$ (C) $A^{30} = A^{25}$ (D) $A^{30} + A^{25} + A = I$

Open solution

Q34. MCQ (Single Correct Answer)

Two dice are thrown independently. Let A be the event that the number appeared on the 1st die is less than the number appeared on the 2nd die, B be the event that the number appeared on the 1st die is even and that on the second die is odd, and C be the event that the number appeared on the 1st die is odd and that on the 2nd is even. Then :

(A) A and B are mutually exclusive

(B) the number of favourable cases of the events A, B and C are 15, 6 and 6 respectively

(C) B and C are independent

(D) the number of favourable cases of the event $(A \cup B) \cap C$ is 6

Open solution

Q35. MCQ (Single Correct Answer)

Let $\vec{a} = 2\hat{i} - 7\hat{j} + 5\hat{k}, \vec{b} = \hat{i} + \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$ and $\vec{r} \cdot \vec{b} = 0$, then $|\vec{r}|$ is equal to :

(A) $\frac{11}{7}$ (B) $\frac{11}{5}\sqrt{2}$ (C) $\frac{\sqrt{914}}{7}$ (D) $\frac{11}{7}\sqrt{2}$

Open solution

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Q36. MCQ (Single Correct Answer)

Let $S = \left\{x \in R : 0 < x < 1 \text{ and } 2\tan^{-1}\left(\frac{1-x}{1+x}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right\}$.

If $\mathrm{n(S)}$ denotes the number of elements in S then :

(A) $\mathrm{n(S)} = 0$

(B) $\mathrm{n(S)} = 2$ and only one element in S is less than $\frac{1}{2}$.

(C) $\mathrm{n(S)} = 1$ and the element in S is more than $\frac{1}{2}$.

(D) $\mathrm{n(S)} = 1$ and the element in S is less than $\frac{1}{2}$.

Open solution

Q37. MCQ (Single Correct Answer)

Let $f : \mathbb{R} - \{0, 1\} \to \mathbb{R}$ be a function such that $f(x) + f\left(\frac{1}{1-x}\right) = 1 + x$. Then $f(2)$ is equal to

(A) $\frac{9}{4}$ (B) $\frac{7}{4}$ (C) $\frac{7}{3}$ (D) $\frac{9}{2}$

Open solution

Q38. MCQ (Single Correct Answer)

Let $\alpha x = \exp\left(x^\beta y^\gamma\right)$ be the solution of the differential equation $2x^2y\,dy - \left(1 - xy^2\right)dx = 0, x > 0, y(2) = \sqrt{\log_e 2}$. Then $\alpha + \beta - \gamma$ equals :

(A) 1 (B) 0 (C) 3 (D) -1

Open solution

Q39. Numerical

If the x-intercept of a focal chord of the parabola $y^2 = 8x + 4y + 4$ is 3, then the length of this chord is equal to _____.

Open solution

Q40. Numerical

If $\int_0^\pi \frac{5^{\cos x}(1+\cos x \cos 3x+\cos^2 x+\cos^3 x \cos 3x)dx}{1+5^{\cos x}} = \frac{k\pi}{16}$, then k is equal to _____.

Open solution

Q41. Numerical

Let the sixth term in the binomial expansion of $\left(\sqrt{2^{\log_2(10-3^x)}} + \sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$ in the increasing powers of $2^{(x-2)\log_2 3}$, be 21 . If the binomial coefficients of the second, third and fourth terms in the expansion are respectively the first, third and fifth terms of an A.P., then the sum of the squares of all possible values of x is _____.

Open solution

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Q42. Numerical

The sum of the common terms of the following three arithmetic progressions.

$3, 7, 11, 15, \ldots, 399$,

$2, 5, 8, 11, \ldots, 359$ and

$2, 7, 12, 17, \ldots, 197$,

is equal to _____.

Open solution

Q43. Numerical

Number of integral solutions to the equation $x + y + z = 21$, where $x \geq 1, y \geq 3, z \geq 4$, is equal to _____.

Open solution

Q44. Numerical

If the term without x in the expansion of $\left(x^{\frac{2}{3}} + \frac{\alpha}{x^3}\right)^{22}$ is 7315 , then $|\alpha|$ is equal to _____.

Open solution

Q45. Numerical

The total number of six digit numbers, formed using the digits 4, 5, 9 only and divisible by 6, is _____.

Open solution

Physics 28 questions

Q46. MCQ (Single Correct Answer)

Given below are two statements : One is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$.

Assertion A : For measuring the potential difference across a resistance of 600Ω, the voltmeter with resistance 1000Ω will be preferred over voltmeter with resistance 4000Ω.

Reason R : Voltmeter with higher resistance will draw smaller current than voltmeter with lower resistance.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) $\mathbf{A}$ is not correct but $\mathbf{R}$ is correct

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is correct but $\mathbf{R}$ is not correct

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

Open solution

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Q47. MCQ (Single Correct Answer)

The escape velocities of two planets A and B are in the ratio $1:2$. If the ratio of their radii respectively is $1:3$, then the ratio of acceleration due to gravity of planet A to the acceleration of gravity of planet B will be :

Open solution

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{3}{2}$

Q48. MCQ (Single Correct Answer)

The ratio of average electric energy density and total average energy density of electromagnetic wave is :

Open solution

(A) 1 (B) 3 (C) 2 (D) $\frac{1}{2}$

Q49. MCQ (Single Correct Answer)

Equivalent resistance between the adjacent corners of a regular n-sided polygon of uniform wire of resistance R would be :

Open solution

(A) $\frac{(\mathrm{n}-1)\mathrm{R}}{\mathrm{n}^2}$ (B) $\frac{n^2R}{n-1}$
(C) $\frac{(n-1)R}{(2n-1)}$ (D) $\frac{(n-1)R}{n}$

Q50. MCQ (Single Correct Answer)

A coil is placed in magnetic field such that plane of coil is perpendicular to the direction of magnetic field. The magnetic flux through a coil can be changed :

Open solution

A. By changing the magnitude of the magnetic field within the coil.

B. By changing the area of coil within the magnetic field.

C. By changing the angle between the direction of magnetic field and the plane of the coil.

D. By reversing the magnetic field direction abruptly without changing its magnitude.

Choose the most appropriate answer from the options given below :

(A) A, B and D only (B) A, B and C only (C) A and B only (D) A and C only

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Q51. MCQ (Single Correct Answer)

Open solution

Given below are two statements: One is labeled as Assertion A and the other is labeled as Reason R.

Assertion A : Two metallic spheres are charged to the same potential. One of them is hollow and another is solid, and both have the same radii. Solid sphere will have lower charge than the hollow one.

Reason R : Capacitance of metallic spheres depends on the radii of spheres

In light of the above statements, choose the correct answer from the options given below.

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(D) $\mathbf{A}$ is true but $\mathbf{R}$ is false

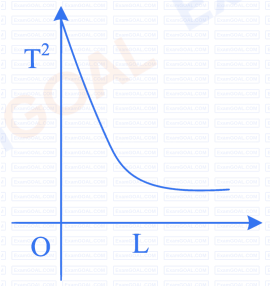
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Q52. MCQ (Single Correct Answer)

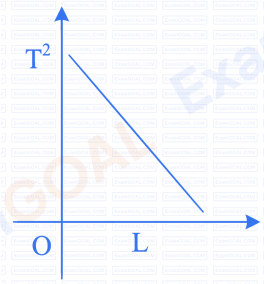
Choose the correct length (L) versus square of the time period (T^2) graph for a simple pendulum executing simple harmonic motion.

Open solution

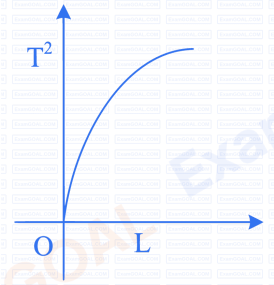
(A)



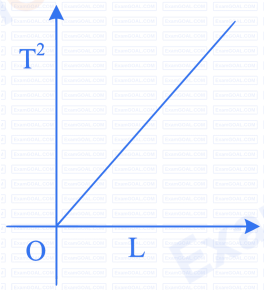
(B)



(C)



(D)



Q53. MCQ (Single Correct Answer)

If the velocity of light c, universal gravitational constant G and Planck's constant h are chosen as fundamental quantities. The dimensions of mass in the new system is :

Open solution

(A) $\left[h^1 c^1 G^{-1}\right]$

(B) $\left[h^{-1/2} c^{1/2} G^{1/2}\right]$

(C) $\left[h^{1/2} c^{1/2} G^{-1/2}\right]$

(D) $\left[h^{1/2} c^{-1/2} G^1\right]$

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Q54. MCQ (Single Correct Answer)

Open solution

The Young's modulus of a steel wire of length $6\ \mathrm{m}$ and cross-sectional area $3\ \mathrm{mm}^2$, is $2 \times 10^{11}\ \mathrm{N/m}^2$. The wire is suspended from its support on a given planet. A block of mass $4\ \mathrm{kg}$ is attached to the free end of the wire. The acceleration due to gravity on the planet is $\frac{1}{4}$ of its value on the earth. The elongation of wire is (Take g on the earth $= 10\ \mathrm{m/s}^2$) :

(A) 0.1 cm (B) 1 cm (C) 0.1 mm (D) 1 mm

Q55. MCQ (Single Correct Answer)

Open solution

Choose the correct statement about Zener diode :

(A) It works as a voltage regulator in reverse bias and behaves like simple pn junction diode in forward bias.

(B) It works as a voltage regulator in both forward and reverse bias.

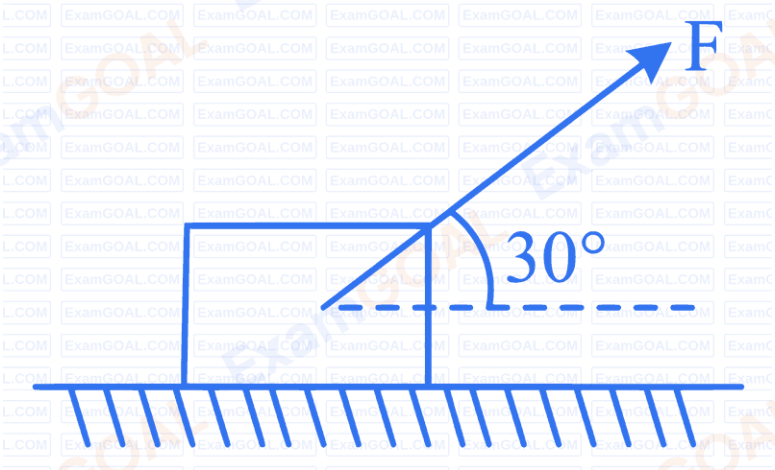
(C) It works as a voltage regulator only in forward bias.

(D) It works as a voltage regulator in forward bias and behaves like simple pn junction diode in reverse bias.

Q56. MCQ (Single Correct Answer)

Open solution

As shown in the figure a block of mass 10 kg lying on a horizontal surface is pulled by a force F acting at an angle 30°, with horizontal. For $\mu_s = 0.25$, the block will just start to move for the value of F : [Given $g = 10\ \mathrm{ms}^{-2}$]



(A) 25.2 N (B) 35.7 N (C) 20 N (D) 33.3 N

Q57. MCQ (Single Correct Answer)

Open solution

Two objects A and B are placed at 15 cm and 25 cm from the pole in front of a concave mirror having radius of curvature 40 cm. The distance between images formed by the mirror is _____.

(A) 60 cm (B) 40 cm (C) 160 cm (D) 100 cm

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Q58. MCQ (Single Correct Answer)

Open solution

For a body projected at an angle with the horizontal from the ground, choose the correct statement.

(A) Gravitational potential energy is maximum at the highest point.

(B) The vertical component of momentum is maximum at the highest point.

(C) The horizontal component of velocity is zero at the highest point.

(D) The Kinetic Energy (K.E.) is zero at the highest point of projectile motion.

Q59. MCQ (Single Correct Answer)

Open solution

As shown in the figure, a long straight conductor with semicircular arc of radius $\frac{\pi}{10}$m is carrying current $\mathrm{I} = 3\mathrm{A}$. The magnitude of the magnetic field, at the center O of the arc is :

(The permeability of the vacuum $= 4\pi \times 10^{-7}\ \mathrm{NA}^{-2}$)



(A) $4\mu\mathrm{T}$ (B) $3\mu\mathrm{T}$ (C) $6\mu\mathrm{T}$ (D) $1\mu\mathrm{T}$

Q60. MCQ (Single Correct Answer)

Open solution

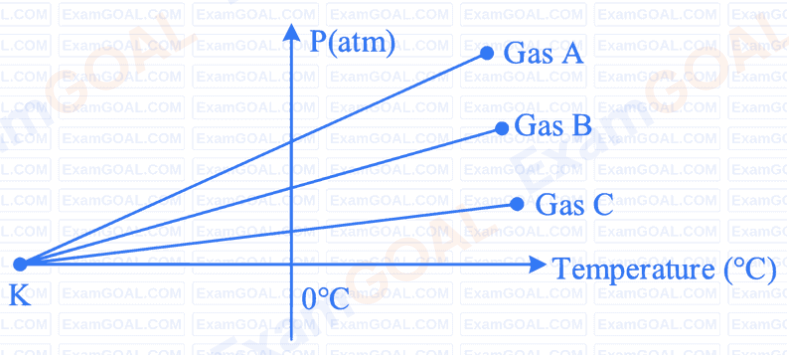
The threshold frequency of a metal is f_0. When the light of frequency $2f_0$ is incident on the metal plate, the maximum velocity of photoelectrons is v_1. When the frequency of incident radiation is increased to $5\mathrm{f}_0$, the maximum velocity of photoelectrons emitted is v_2. The ratio of v_1 to v_2 is :

(A) $\frac{v_1}{v_2} = \frac{1}{2}$ (B) $\frac{v_1}{v_2} = \frac{1}{16}$ (C) $\frac{v_1}{v_2} = \frac{1}{4}$ (D) $\frac{v_1}{v_2} = \frac{1}{8}$

Q61. MCQ (Single Correct Answer)

Open solution

For three low density gases A, B, C pressure versus temperature graphs are plotted while keeping them at constant volume, as shown in the figure.



The temperature corresponding to the point 'K' is :

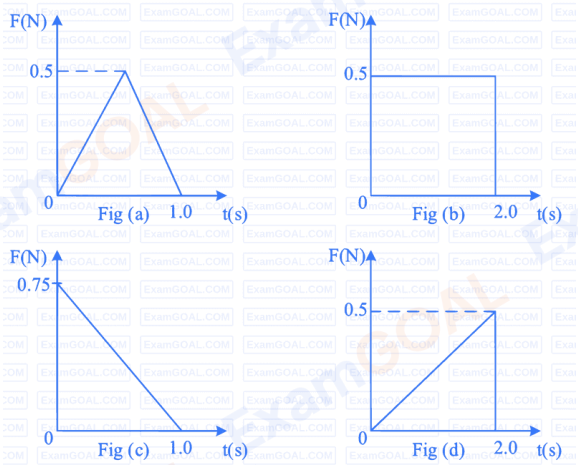
(A) $-273^\circ\mathrm{C}$ (B) $-373^\circ\mathrm{C}$ (C) $-100^\circ\mathrm{C}$ (D) $-40^\circ\mathrm{C}$

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Q62. MCQ (Single Correct Answer)

Open solution

Figures (a), (b), (c) and (d) show variation of force with time.



The impulse is highest in figure.

(A) Fig (c) (B) Fig (d) (C) Fig (a) (D) Fig (b)

Q63. MCQ (Single Correct Answer)

Open solution

An electron of a hydrogen like atom, having $Z = 4$, jumps from 4th energy state to 2nd energy state. The energy released in this process, will be :

(Given Rch = $13.6\ \text{eV}$)

Where R = Rydberg constant

c = Speed of light in vacuum

h = Planck's constant

(A) $10.5\ \text{eV}$ (B) $40.8\ \text{eV}$ (C) $13.6\ \text{eV}$ (D) $3.4\ \text{eV}$

Q64. Numerical

Open solution

A square shaped coil of area $70\ \text{cm}^2$ having 600 turns rotates in a magnetic field of $0.4\ \text{wbm}^{-2}$, about an axis which is parallel to one of the sides of the coil and perpendicular to the direction of field. If the coil completes 500 revolutions in a minute, the instantaneous emf when the plane of the coil is inclined at 60° with the field, will be _____ V. (Take $\pi = \frac{22}{7}$)

Q65. Numerical

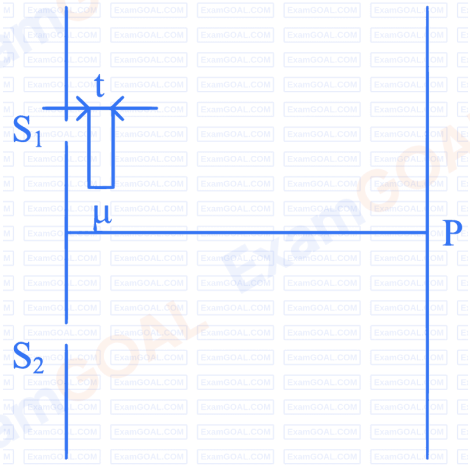
Open solution

A block is fastened to a horizontal spring. The block is pulled to a distance $x = 10\ \text{cm}$ from its equilibrium position (at $x = 0$) on a frictionless surface from rest. The energy of the block at $x = 5\ \text{cm}$ is $0.25\ \text{J}$. The spring constant of the spring is _____ Nm^{-1}

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Q66. Numerical

As shown in the figure, in Young's double slit experiment, a thin plate of thickness $t = 10\mu m$ and refractive index $\mu = 1.2$ is inserted in front of slit S_1. The experiment is conducted in air $(\mu = 1)$ and uses a monochromatic light of wavelength $\lambda = 500\,\text{nm}$. Due to the insertion of the plate, central maxima is shifted by a distance of $x\beta_0$. β_0 is the fringe-width before the insertion of the plate. The value of the x is _____.



Open solution

Q67. Numerical

Nucleus A having $Z = 17$ and equal number of protons and neutrons has $1.2\,\text{MeV}$ binding energy per nucleon.

Another nucleus B of $Z = 12$ has total 26 nucleons and $1.8\,\text{MeV}$ binding energy per nucleons.

The difference of binding energy of B and A will be _____ MeV.

Open solution

Q68. Numerical

Moment of inertia of a disc of mass 'M' and radius 'R' about any of its diameter is $\frac{MR^2}{4}$. The moment of inertia of this disc about an axis normal to the disc and passing through a point on its edge will be, $\frac{x}{2}$ MR^2. The value of x is _____.

Open solution

Q69. Numerical

The surface of water in a water tank of cross section area $750\,\text{cm}^2$ on the top of a house is h m above the tap level. The speed of water coming out through the tap of cross section area $500\,\text{mm}^2$ is $30\,\text{cm/s}$. At that instant, $\frac{dh}{dt}$ is $x \times 10^{-3}$ m/s. The value of x will be _____.

Open solution

Q70. Numerical

A force $\text{F} = (5 + 3y^2)$ acts on a particle in the y-direction, where F is in newton and y is in meter. The work done by the force during a displacement from $y = 2$ m to $y = 5$ m is _____ J.

Open solution

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Q71. Numerical

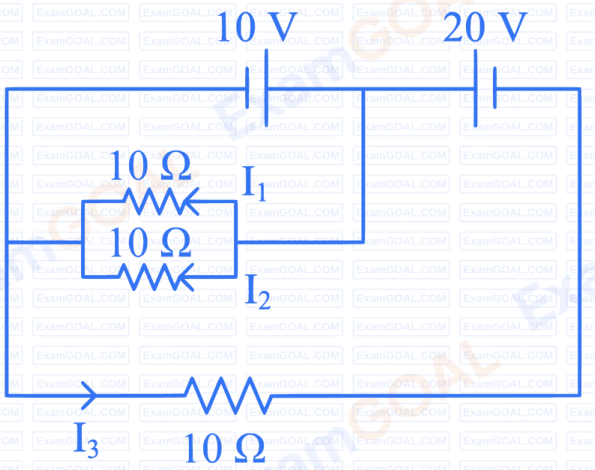
For a train engine moving with speed of $20\ \mathrm{ms}^{-1}$, the driver must apply brakes at a distance of 500 m before the station for the train to come to rest at the station. If the brakes were applied at half of this distance, the train engine would cross the station with speed $\sqrt{x}\ \mathrm{ms}^{-1}$. The value of x is _____.

(Assuming same retardation is produced by brakes)

Open solution

Q72. Numerical

In the given circuit, the value of $\left|\frac{I_1+I_3}{I_2}\right|$ is _____



Open solution

Q73. Numerical

A cubical volume is bounded by the surfaces $\mathrm{x} = 0, x = \mathrm{a}, y = 0, y = \mathrm{a}, \mathrm{z} = 0, z = \mathrm{a}$. The electric field in the region is given by $\vec{\mathrm{E}} = \mathrm{E}_0 x\hat{i}$. Where $\mathrm{E}_0 = 4 \times 10^4\ \mathrm{NC}^{-1}\ \mathrm{m}^{-1}$. If $\mathrm{a} = 2\ \mathrm{cm}$, the charge contained in the cubical volume is $\mathrm{Q} \times 10^{-14}\mathrm{C}$. The value of Q is _____.

(Take $\epsilon_0 = 9 \times 10^{-12}\ \mathrm{C}^2/\mathrm{Nm}^2$)

Open solution

Out of syllabus 17 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q74. MCQ (Single Correct Answer) · **Chemistry**

The industrial activity held least responsible for global warming is :

(A) manufacturing of cement

(B) Electricity generation in thermal power plants

(C) steel manufacturing

(D) Industrial production of urea

Open solution

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Q75. MCQ (Single Correct Answer) · **Chemistry**

The starting material for convenient preparation of deuterated hydrogen peroxide (D_2O_2) in :

(A) BaO_2

(B) 2-ethylanthraquinol

(C) BaO

(D) $K_2S_2O_8$

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

The correct order of bond enthalpy ($kJ\ mol^{-1}$) is :

(A) $C-C > Si-Si > Sn-Sn > Ge-Ge$

(B) $Si-Si > C-C > Ge-Ge > Sn-Sn$

(C) $C-C > Si-Si > Ge-Ge > Sn-Sn$

(D) $Si-Si > C-C > Sn-Sn > Ge-Ge$

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Gypsum is used for making fireproof wall boards.

Reason (R) : Gypsum is unstable at high temperatures.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(B) Both (A) and (R) are correct and (R) is the correct explanation of (A)

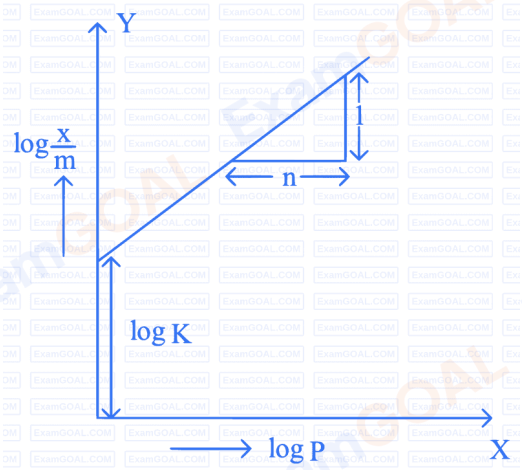
(C) (A) is correct but (R) is not correct

(D) (A) is not correct but (R) is correct

Open solution

Q78. MCQ (Single Correct Answer) · **Chemistry**

In figure, a straight line is given for Freundrich Adsorption ($y = 3x + 2.505$). The value of $\frac{1}{n}$ and $\log K$ are respectively.



(A) 0.3 and log 2.505

(B) 3 and 2.505

(C) 3 and 0.7033

(D) 0.3 and 0.7033

Open solution

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Q79. Numerical · **Chemistry**

Among the following, the number of tranquilizer/s is/are _____.

A. Chlordiazepoxide

B. Veronal

C. Valium

D. Salvarsan

Open solution

Q80. Numerical · **Chemistry**

Among the following compounds, the number of those present in copper matte is _____.

A. $\mathrm{CuCO_3}$

B. $\mathrm{Cu_2S}$

C. $\mathrm{Cu_2O}$

D. FeO

Open solution

Q81. Numerical · **Chemistry**

A metal M crystallizes into two lattices :- face centred cubic (fcc) and body centred cubic (bcc) with unit cell edge length of 2.0 and $2.5\ \overset{o}{A}$ respectively. The ratio of densities of lattices fcc to bcc for the metal M is _____. (Nearest integer)

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let the plane P pass through the intersection of the planes $2x + 3y - z = 2$ and $x + 2y + 3z = 6$, and be perpendicular to the plane $2x + y - z + 1 = 0$. If d is the distance of P from the point $(-7, 1, 1)$, then d^2 is equal to :

(A) $\frac{250}{83}$ (B) $\frac{250}{82}$ (C) $\frac{15}{53}$ (D) $\frac{25}{83}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let $\mathrm{P}\,(x_0, y_0)$ be the point on the hyperbola $3x^2 - 4y^2 = 36$, which is nearest to the line $3x + 2y = 1$. Then $\sqrt{2}\,(y_0 - x_0)$ is equal to :

(A) 3 (B) -9 (C) -3 (D) 9

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Which of the following statements is a tautology?

(A) $\mathrm{p} \vee (\mathrm{p} \wedge \mathrm{q})$ (B) $(\mathrm{p} \wedge (\mathrm{p} \to \mathrm{q})) \to \sim \mathrm{q}$

(C) $\mathrm{p} \to (\mathrm{p} \wedge (\mathrm{p} \to \mathrm{q}))$ (D) $(\mathrm{p} \wedge \mathrm{q}) \to (\sim (\mathrm{p}) \to \mathrm{q})$

Open solution

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Q85. MCQ (Single Correct Answer) · **Mathematics**

The sum $\sum_{n=1}^{\infty} \frac{2n^2+3n+4}{(2n)!}$ is equal to :

(A) $\frac{11e}{2} + \frac{7}{2e}$

(B) $\frac{13e}{4} + \frac{5}{4e} - 4$

(C) $\frac{11e}{2} + \frac{7}{2e} - 4$

(D) $\frac{13e}{4} + \frac{5}{4e}$

Open solution

Q86. Numerical · **Mathematics**

The point of intersection C of the plane $8x + y + 2z = 0$ and the line joining the points $\mathrm{A}(-3, -6, 1)$ and $\mathrm{B}(2, 4, -3)$ divides the line segment AB internally in the ratio $\mathrm{k} : 1$. If $\mathrm{a}, \mathrm{b}, \mathrm{c}(|\mathrm{a}|, |\mathrm{b}|, |\mathrm{c}|$ are coprime) are the direction ratios of the perpendicular from the point C on the line $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$, then $|\mathrm{a} + \mathrm{b} + \mathrm{c}|$ is equal to _____.

Open solution

Q87. Numerical · **Mathematics**

Let $\alpha x + \beta y + \gamma z = 1$ be the equation of a plane passing through the point $(3, -2, 5)$ and perpendicular to the line joining the points $(1, 2, 3)$ and $(-2, 3, 5)$. Then the value of $\alpha\beta\gamma$ is equal to _____.

Open solution

Q88. Numerical · **Mathematics**

The line $x = 8$ is the directrix of the ellipse $\mathrm{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with the corresponding focus $(2, 0)$. If the tangent to E at the point P in the first quadrant passes through the point $(0, 4\sqrt{3})$ and intersects the x-axis at Q, then $(3\mathrm{PQ})^2$ is equal to _____.

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

In an amplitude modulation, a modulating signal having amplitude of X Volt is superimposed with a carrier signal of amplitude Y Volt in first case. Then, in second case, the same modulating signal is superimposed with different carrier signal of amplitude 2Y Volt. The ratio of modulation index in the two cases respectively will be :

(A) 2 : 1

(B) 1 : 1

(C) 4 : 1

(D) 1 : 2

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

A Carnot engine operating between two reservoirs has efficiency $\frac{1}{3}$. When the temperature of cold reservoir is raised by x, its efficiency decreases to $\frac{1}{6}$. The value of x, if the temperature of hot reservoir is 99°C, will be :

(A) 66 K

(B) 62 K

(C) 16.5 K

(D) 33 K

Open solution

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Answer Key

Chemistry

1. **B**	2. **B**	3. **D**	4. **B**	5. **D**
6. **D**	7. **D**	8. **A**	9. **A, B**	10. **D**
11. **D**	12. **D**	13. **C**	14. **A**	15. **A**
16. **13039**	17. **6**	18. **372**	19. **1006**	20. **139**
21. **6**	22. **75**			

Mathematics

23. **D**	24. **C**	25. **C**	26. **A**	27. **B**
28. **D**	29. **A**	30. **A**	31. **C**	32. **C**
33. **B**	34. **D**	35. **D**	36. **D**	37. **A**
38. **A**	39. **16**	40. **13**	41. **4**	42. **321**
43. **105**	44. **1**	45. **81**		

Physics

46. **A**	47. **C**	48. **D**	49. **A**	50. **B**
51. **C**	52. **D**	53. **C**	54. **C**	55. **A**
56. **A**	57. **C**	58. **A**	59. **B**	60. **A**
61. **A**	62. **D**	63. **B**	64. **44**	65. **67**
66. **4**	67. **6**	68. **3**	69. **2**	70. **132**
71. **200**	72. **2**	73. **288**		

Out of syllabus

74. **D**	75. **D**	76. **C**	77. **A**	78. **B**
79. **3**	80. **1**	81. **4**	82. **A**	83. **B**
84. **D**	85. **B**	86. **10**	87. **6**	88. **39**
89. **A**	90. **B**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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